

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2020

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2020 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. (a) Expand and simplify $(4x - 1)(x + 1)$. [3]<<
>>(b) Factorise completely $8x^2 + 8x$. [2]<<
>>(c) Solve $4(x - 1) = 12$. [3]<<
>>(d) Simplify $(x^2 - 4)/(x - 2)$. [2]
[10]

2. AB is parallel to CD. Transversal PQ meets AB at X and CD at Y.<<
>>(a) Angle $AXY = 61^\circ$. Find the corresponding angle at Y. Give a reason. [2]<<
>>(b) Find the co-interior (allied) angle to 61° . Give a reason. [2]<<
>>(c) In triangle PQR, $P = 61^\circ$ and $Q = 37^\circ$. Calculate R. [2]<<
>>(d) State whether triangle PQR is acute, right-angled or obtuse. Give a reason. [2] **[8]**

3. A closed cylinder has radius 7 cm and height 8 cm. Take $\pi = 22/7$.<<
>>(a) Calculate the volume. [3]<<
>>(b) Calculate the curved surface area. [3]<<
>>(c) Calculate the total surface area. [2] **[8]**

4. The line L has equation $y = -2x + -5$.<<
>>(a) Write down the gradient of L and the coordinates of the y-intercept. [2]<<
>>(b) Find the coordinates of the point where L meets the x-axis. [2]<<
>>(c) Find the gradient of the line joining A(3, 3) and B(7, 7). [2]<<
>>(d) Write the equation of the line parallel to L that passes through (0, 1). [2]
[8]

5. The table shows values of x with frequencies f:<<
>>x=1, f=2, x=3, f=7, x=3, f=3, x=5, f=6, x=5, f=5.<<
>>(a) State the modal value of x. [1]<<
>>(b) Calculate Σf and Σfx . [3]<<
>>(c) Calculate the mean of x. [3]<<
>>(d) Find the range of x. [2] **[9]**

6. $A = \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}$ and column vector $u = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.
[3] (a) Calculate Au .
[2] (b) Find $\det A$.
[2] (c) Describe fully the transformation represented by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.
[2] (d) Find the image of the point $(2, 3)$ under a reflection in the y -axis. [2] **[9]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** (a) Factorise $x^2 - 7x + 10$. [3] (b) Hence solve $x^2 - 7x + 10 = 0$. [2] (c) Write down the roots of $(x - 2)(x - 5) = 0$. [1] (d) Sketch $y = (x - 2)(x - 5)$, showing the intercepts with both axes. [3] (e) State the coordinates of the turning point. [3] **[12]**
- 8.** In triangle ABC, $\hat{C} = 90^\circ$, $\hat{A} = 30^\circ$ and AC = 12 cm (hypotenuse). (a) Write down $\sin 30^\circ$ and $\cos 60^\circ$. [2] (b) Calculate BC (opposite 30°). [3] (c) Calculate AB (adjacent to 30°). [3] (d) Show that $\tan 30^\circ = 1/\sqrt{3}$ and hence find BC/AB. [4] **[12]**
- 9.** O is the centre of a circle. A, B and C lie on the circumference. AB is a diameter. Angle BAC = 36° . (a) Find angle ACB. Give a reason. [3] (b) Find angle ABC. [2] (c) Find angle BOC (angle at the centre standing on arc BC). Give a reason. [4] (d) Point D lies on the remaining circumference. Find angle BDC. Give a reason. [3] **[12]**
- 10.** The position vectors of A and B are $\mathbf{a} = (5 ; 5)$ and $\mathbf{b} = (2 ; 2)$. M is the midpoint of AB. (a) Find vector AB. [3] (b) Find $|\mathbf{AB}|$. [3] (c) Find the position vector of M. [3] (d) Show that $\mathbf{AM} = \frac{1}{2} \mathbf{AB}$. [3] **[12]**
- 11.** 60 candidates sat a Mathematics paper marked out of 50. A cumulative frequency curve (ogive) is drawn. (a) State the median position on the cumulative-frequency axis. [2] (b) Describe how to read the median mark from the ogive. [3] (c) The lower quartile is at position $n/4$ and the upper quartile at $3n/4$. Write these positions as numbers. [3] (d) Define the interquartile range and state one advantage of the IQR over the range. [4] **[12]**
- 12.** Triangle PQR has P(2, 4), Q(5, 4) and R(2, 7). (a) P' is the image of P under reflection in the y-axis. Write down the coordinates of P'. [2] (b) Q is translated by the vector $(1 ; -2)$. Find the image Q''. [3] (c) The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ represents a transformation T. Describe T fully. [4] (d) Find T(Q), the image of Q under T. [3] **[12]**
- 13.** (a) y varies directly as x . When $x = 3$, $y = 12$. (i) Find the equation connecting y and x . [3] (ii) Find y when $x = 12$. [2] (b) z varies inversely as x . When $x = 3$, $z = 4$. Let this value of z be z_0 . (i) Find z when $x = 6$. [4] (ii) Describe what happens to z when x is doubled. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).